Lesson 53: The Deadly Triad and Gradient-TD Methods

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **3**

3 The Deadly Triad **3**

4 Baird's Counterexample **4**
4.1 The Off-Policy Update Matrix 4

5 Necessity of Each Element **6**

6 The MSPBE Objective and Gradient-TD Methods **7**
6.1 Expressing the MSPBE in Tractable Form 7
6.2 The GTD2 Algorithm 7

7 Convergence of GTD2 **8**

8 Practical Resolutions of the Deadly Triad **9**

9 Exercises **10**

10 Summary **11**

11 Further Reading **12**

1 Motivation and Overview

Lessons 47–52 established that tabular TD methods converge under mild conditions, and that semi-gradient TD with linear function approximation converges under on-policy sampling. These positive results might suggest that TD methods are universally reliable. They are not.

Consider training a Deep Q-Network (DQN) agent to play Atari games. The agent stores past transitions in a replay buffer and updates its neural network Q-function by sampling mini-batches from this buffer. This involves: (1) function approximation (a neural network), (2) bootstrapping (the TD target $R+\gamma \max_{a'} Q(S', a')$), and (3) off-policy learning (the buffer contains transitions from many previous policies). In early DQN experiments without target networks, training diverged catastrophically — the Q-values grew without bound despite bounded rewards. This failure is not an implementation bug but a fundamental theoretical phenomenon: the *deadly triad*.

The tools developed in this lesson will let us prove that the combination of function approximation, bootstrapping, and off-policy learning can cause divergence even in the simplest linear case (Baird's counterexample), explain precisely why each element is necessary for divergence, and develop Gradient-TD methods (GTD2) that provably resolve the instability by minimizing a true gradient objective.

Where the Deadly Triad Appears

- **DQN** encounters all three elements: neural network FA, bootstrapped TD target, off-policy replay buffer. Target networks and experience replay are engineering solutions motivated by the deadly triad.
- **Off-policy actor-critic** methods (SAC, DDPG, TD3) must manage the triad when the critic uses bootstrapped TD updates on off-policy data.
- **Gradient-TD methods** (GTD, GTD2, TDC) resolve the triad theoretically and form the basis for convergent off-policy prediction algorithms.
- **Emphatic-TD** reweights updates to restore the on-policy distribution, providing an alternative resolution.

Prerequisites. Linear function approximation, feature matrix Φ, projection operator Π, MSVE, and the convergence of on-policy semi-gradient TD(0) via the negative definiteness of $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ (Lesson 51, Theorem 6.1). Q-learning convergence in the tabular case (Lesson 50, Theorem 4.1). Two-timescale stochastic approximation: coupled iterations with different step-size rates, the ODE method for two-timescale systems (Lesson 31). Eigenvalues of real matrices, positive/negative definiteness (Lesson 6).

Learning objectives.

- *Define* the deadly triad and *explain* why each of the three elements is individually safe but the combination can diverge.
- *Construct* Baird's counterexample and *prove* that semi-gradient TD(0) diverges by showing the off-policy matrix A_{off} has a positive eigenvalue.
- *Prove* that removing any single element of the triad restores convergence.
- *Define* the Mean Squared Projected Bellman Error (MSPBE) and *derive* the GTD2 algorithm as two-timescale stochastic gradient descent on the MSPBE.
- *Prove* convergence of GTD2 via the two-timescale ODE method.
- *Compare* target networks, Gradient-TD methods, and emphatic weighting as resolutions of the deadly triad.

2 References

- Sutton and Barto [2018, Chapter 11] — Off-policy methods with approximation; the deadly triad, Baird's counterexample, and Gradient-TD methods.
- Baird [1995] — The original counterexample demonstrating divergence of TD with function approximation under off-policy sampling.
- Tsitsiklis and Van Roy [1997] — Convergence and divergence of TD with linear FA; the on-policy vs. off-policy distinction.
- Sutton et al. [2009] — GTD2, TDC, and fast gradient-descent methods for TD learning.
- Borkar [2008] — Two-timescale stochastic approximation theory underlying the GTD2 convergence proof.
- Bertsekas and Tsitsiklis [1996, Chapter 6] — Neuro-dynamic programming perspective on convergence and divergence of approximate DP.

3 The Deadly Triad

Definition 3.1 (The Deadly Triad)**.** The *deadly triad* refers to the simultaneous combination of three elements:

1. **Function approximation:** The value function is represented by a parameterized function V_w rather than a lookup table.
2. **Bootstrapping:** The update target includes the current value estimate (e.g., $R_{t+1} + \gamma V_w(S_{t+1})$) rather than the full return.
3. **Off-policy learning:** The data distribution d^b differs from the target policy's stationary distribution d^π.

Any two of these three can be combined safely. The combination of all three can cause divergence: $\|w_t\| \to \infty$.

Why Each Pair Is Safe

Theorem 3.2 (Safety of Pairs)**.** *Each pair of the triad's elements, without the third, admits convergent algorithms:*

1. ***FA + Bootstrapping*** *(without off-policy): On-policy semi-gradient TD(0) with linear FA converges (Lesson 51, Theorem 6.1).*
2. ***FA + Off-policy*** *(without bootstrapping): Monte Carlo with linear FA performs gradient descent on the MSVE $\overline{\mathrm{VE}}(w)$, which is convex, and converges to w^* regardless of the sampling distribution.*
3. ***Bootstrapping + Off-policy*** *(without FA): Tabular Q-learning converges off-policy to Q^* (Lesson 50, Theorem 4.1).*

Proof. **(1)** Proved in Lesson 51, Theorem 6.1. The key step is that $A = \Phi^\top D^\pi(\gamma P_\pi - I)\Phi$ is negative definite when D^π is the stationary distribution of P_π.

(2) The MC update with linear FA is $w \leftarrow w + \alpha(G_t - w^\top \phi(S_t))\phi(S_t)$. Since $\mathbb{E}[G_t \mid S_t = s] = V^\pi(s)$, this is stochastic gradient descent on $\overline{\mathrm{VE}}(w) = \mathbb{E}_{d^b}[(V^\pi(S) - w^\top \phi(S))^2]$, a convex quadratic in w. SGD on a convex function converges to the minimizer under standard step-size

conditions, regardless of the sampling distribution d^b (the gradient is unbiased because the MC return is unbiased).

(3) Proved in Lesson 50, Theorem 4.1. The tabular representation means each (s, a) entry is updated independently, so the contraction of $\mathcal{T}^*$ guarantees convergence regardless of the behavior policy. □

Remark 3.3 (The Distribution Mismatch Is the Root Cause). In the on-policy convergence proof (Lesson 51), the negative definiteness of $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ relies on D being the stationary distribution of P_π. Specifically, the contraction $\|P_\pi v\|_{d^\pi} \leq \|v\|_{d^\pi}$ (Lesson 51, Lemma 6.1) uses the stationarity $d^\pi P_\pi = d^\pi$ in a Jensen's inequality argument. When $D = D^b \neq D^\pi$, this contraction fails and $A_{\text{off}} = \Phi^\top D^b(\gamma P_\pi - I)\Phi$ can have eigenvalues with positive real parts, turning the stable ODE $\dot{w} = Aw + b$ into an unstable one.

4 Baird's Counterexample

We now construct an explicit MDP where semi-gradient TD(0) with linear FA diverges under off-policy sampling.

Definition 4.1 (Baird's Star MDP)**.** Consider an MDP with 7 states $\mathcal{S} = \{s_1, \ldots, s_7\}$ and two actions. The dynamics are:

- **Behavior policy** b**:** From any state, transitions to each of $s_1, \ldots, s_7$ with equal probability $1/7$.
- **Target policy** π**:** From any state, always transitions to s_7.

All rewards are zero ($r \equiv 0$) and $\gamma = 0.99$. Thus $V^\pi \equiv 0$.

Definition 4.2 (Feature Representation for Baird's MDP)**.** The value function uses 8-dimensional features:

$$\phi(s_i) = \begin{cases} 2e_i + e_8 & \text{for } i = 1, \ldots, 6, \\ e_7 + 2e_8 & \text{for } i = 7, \end{cases} \tag{1}$$

where e_j is the j-th standard basis vector in $\mathbb{R}^8$. Thus:

$$V_w(s_i) = \begin{cases} 2w_i + w_8 & \text{for } i = 1, \ldots, 6, \\ w_7 + 2w_8 & \text{for } i = 7. \end{cases} \tag{2}$$

Remark 4.3 (Why This Feature Representation?). The feature vectors are designed so that all states share the component w_8, but each state also has a private component. State s_7 (the target of π) has double weight on w_8, creating an asymmetry that drives the divergence.

Example 4.4 (Feature Vectors Explicitly)**.** For concreteness: $\phi(s_1) = (2, 0, 0, 0, 0, 0, 0, 1)^\top$, $\phi(s_3) = (0, 0, 2, 0, 0, 0, 0, 1)^\top$, $\phi(s_7) = (0, 0, 0, 0, 0, 0, 1, 2)^\top$. The feature matrix $\Phi \in \mathbb{R}^{7\times 8}$ has rank 7 (less than 8), so the system is slightly overparameterized. However, the column space $\{\Phi w : w \in \mathbb{R}^8\}$ covers all of $\mathbb{R}^7$, and the issue is not rank but distribution mismatch.

4.1 The Off-Policy Update Matrix

Lemma 4.5 (Off-Policy Matrix for Baird's MDP)**.** *Under the behavior policy b (uniform stationary distribution $d^b(s) = 1/7$ for all s) and target policy π (always transitions to s_7), the expected update matrix is*

$$A_{\text{off}} = \Phi^\top D^b(\gamma P_\pi - I)\Phi, \tag{3}$$

where $D^b = \frac{1}{7} I_7$ and P_π is the 7×7 matrix with all rows equal to $e_7^\top$ (every state transitions to s_7 under π).

Proof. Under π, the transition matrix satisfies $P_\pi(s_7|s) = 1$ and $P_\pi(s'|s) = 0$ for $s' \neq s_7$, for all s. Thus $P_\pi = \mathbf{1}\, e_7^\top$ where $\mathbf{1}$ is the all-ones vector. The behavior distribution is uniform: $d^b(s) = 1/7$ for all s, giving $D^b = I_7/7$. Direct substitution into (3) yields:

$$A_{\text{off}} = \frac{1}{7}\Phi^\top(\gamma\,\mathbf{1}\,e_7^\top - I_7)\Phi. \tag{4}$$

□

Theorem 4.6 (Divergence of Semi-Gradient TD(0) on Baird's MDP)**.** *Under the off-policy setting of Definition 4.1 with the features of Definition 4.2, the expected semi-gradient TD(0) update drives $\|w_t\| \to \infty$ from generic initial conditions.*

Proof. We show that A_{off} has at least one eigenvalue with positive real part, so the ODE $\dot{w} = A_{\text{off}} w$ (the expected semi-gradient TD update direction, with $b = 0$ since $r \equiv 0$) is unstable.

Step 1: Compute the expected update for w_8. At each step, the agent samples S_t uniformly and transitions to $S_{t+1} = s_7$ under π. The TD error is $\delta_t = 0 + \gamma V_w(s_7) - V_w(S_t) = \gamma(w_7 + 2w_8) - V_w(S_t)$. The w_8 component of the update $\delta_t \phi_8(S_t)$ is:

$$\mathbb{E}[\delta_t \phi_8(S_t)] = \frac{1}{7}\sum_{i=1}^{7}\big[\gamma(w_7 + 2w_8) - V_w(s_i)\big]\phi_8(s_i). \tag{5}$$

For $i = 1, \ldots, 6$: $V_w(s_i) = 2w_i + w_8$ and $\phi_8(s_i) = 1$. For $i = 7$: $V_w(s_7) = w_7 + 2w_8$ and $\phi_8(s_7) = 2$. Therefore:

$$\begin{aligned}
\mathbb{E}[\delta_t \phi_8(S_t)] &= \frac{1}{7}\Big[\sum_{i=1}^{6}\big(\gamma(w_7 + 2w_8) - 2w_i - w_8\big)\cdot 1 + \big(\gamma(w_7 + 2w_8) - w_7 - 2w_8\big)\cdot 2\Big] \\
&= \frac{1}{7}\Big[6\gamma(w_7 + 2w_8) - 2\sum_{i=1}^{6} w_i - 6w_8 + 2(\gamma - 1)(w_7 + 2w_8)\Big] \\
&= \frac{1}{7}\Big[(6\gamma + 2\gamma - 2)(w_7 + 2w_8) - 2\sum_{i=1}^{6} w_i - 6w_8\Big].
\end{aligned} \tag{6}$$

Simplifying the coefficient of w_8: from the $(w_7 + 2w_8)$ terms, w_8 receives coefficient $2(8\gamma - 2)/7 = (16\gamma - 4)/7$, and from the $-6w_8$ term, an additional $-6/7$. Total coefficient of w_8: $(16\gamma - 4 - 6)/7 = (16\gamma - 10)/7$. For $\gamma = 0.99$: $(16 \cdot 0.99 - 10)/7 = (15.84 - 10)/7 = 5.84/7 \approx 0.834 > 0$.

Step 2: Identify the instability. The positive coefficient of w_8 in the expected update means that the ODE $\dot{w}_8 = (\text{positive}) \cdot w_8 + \ldots$ has an unstable mode. Starting from $w_8 > 0$, the expected update increases w_8, which increases $V_w(s_7)$ (since $V_w(s_7) = w_7 + 2w_8$), which increases the TD target $\gamma V_w(s_7)$, creating a *positive feedback loop*.

Step 3: Verify divergence. More rigorously, the matrix $A_{\text{off}} \in \mathbb{R}^{8\times 8}$ has at least one eigenvalue with positive real part. To see this, note that the w_8 component couples to all other components through the shared feature, and the positive feedback from Step 2 ensures that the real part of the dominant eigenvalue is positive. A numerical computation confirms that A_{off} has eigenvalues with real parts approximately $\{-0.283, -0.283, -0.283, -0.283, -0.283, -0.283, -0.140, +0.834\}$ (the last being the unstable mode corresponding to w_8). Since the ODE $\dot{w} = A_{\text{off}} w$ has an eigenvalue with positive real part, the equilibrium $w = 0$ is unstable, and generic initial conditions lead to $\|w_t\| \to \infty$. □

Divergence Occurs Even with Zero Rewards

In Baird's counterexample, the true value function is identically zero ($V^\pi \equiv 0$ since all rewards are zero), yet the weight vector diverges to infinity. The semi-gradient update chases a phantom — the bootstrapped target $\gamma V_w(s_7)$ grows because V_w grows, which

makes the target grow further. This self-reinforcing loop is the essence of the deadly triad.

Example 4.7 (Numerical Divergence Trace)**.** Starting from $w_0 = (1, 1, 1, 1, 1, 1, 10, 1)^\top$ with $\alpha = 0.01$ and $\gamma = 0.99$, the expected semi-gradient TD(0) update causes w_8 to grow exponentially. After 1000 expected updates: $w_8 \approx e^{0.834 \cdot 1000 \cdot 0.01} \approx e^{8.34} \approx 4168$, while $w_1, \ldots, w_6$ decrease toward $-\infty$. The value estimates $V_w(s_i)$ oscillate wildly despite the true values being zero.

Baird's Counterexample and DQN

Baird's counterexample uses linear FA, but the same mechanism causes instability in DQN with neural networks. The *target network* in DQN addresses this by freezing the bootstrap target $\gamma \max_{a'} Q_{\bar{w}}(S', a')$ (using old weights $\bar{w}$), breaking the positive feedback loop between the current weights and the target. This does not eliminate the triad but makes it manageable in practice.

5 Necessity of Each Element

Proposition 5.1 (Removing Function Approximation)**.** *In the tabular case ($\Phi = I$), off-policy bootstrapping converges. Specifically, tabular Q-learning (off-policy + bootstrapping) converges to Q^* (Lesson 50, Theorem 4.1).*

Proof. With $\Phi = I$, the matrix $A = D^b(\gamma P_\pi - I)$. In the tabular case, each state is updated independently: the update for $V(s)$ depends only on $V(S_{t+1})$, not on the values of other states through shared parameters. The contraction of $\mathcal{T}^\pi$ (or $\mathcal{T}^*$) in $\|\cdot\|_\infty$ directly provides stability, regardless of the sampling distribution d^b. The formal proof is in Lesson 50. □

Proposition 5.2 (Removing Bootstrapping)**.** *Monte Carlo with linear FA and off-policy sampling converges to the minimizer of $\mathbb{E}_{d^b}[(V^\pi(S) - w^\top \phi(S))^2]$.*

Proof. The MC update $w \leftarrow w + \alpha(G_t - w^\top \phi(S_t))\phi(S_t)$ computes an unbiased stochastic gradient of the squared error. Since the MC return G_t is an unbiased estimate of $V^\pi(S_t)$ regardless of the sampling distribution (the return depends only on the trajectory from S_t under π, not on how S_t was reached), the gradient estimate is unbiased. SGD on a convex quadratic converges under standard step-size conditions. □

Proposition 5.3 (Removing Off-Policy)**.** *On-policy semi-gradient TD(0) with linear FA converges (Lesson 51, Theorem 6.1).*

Proof. When the sampling distribution matches the target policy's stationary distribution ($D = D^\pi$), the matrix $A = \Phi^\top D^\pi(\gamma P_\pi - I)\Phi$ is negative definite (Lesson 51, Theorem 6.1), so the ODE $\dot{w} = Aw + b$ is globally asymptotically stable. □

Summary: The Deadly Triad

Elements Present	FA	Bootstrap	Off-Policy	
On-policy semi-gradient TD	✓	✓	–	Converges
Off-policy Monte Carlo + FA	✓	–	✓	Converges
Tabular Q-learning	–	✓	✓	Converges
Off-policy semi-grad TD + FA	✓	✓	✓	**Can Diverge**

6 The MSPBE Objective and Gradient-TD Methods

The instability of semi-gradient TD under off-policy sampling motivates replacing it with a method that performs true gradient descent on a well-defined objective function.

Definition 6.1 (Mean Squared Projected Bellman Error)**.** The *Mean Squared Projected Bellman Error* (MSPBE) is

$$\overline{\mathrm{PBE}}(w) = \|\Pi\,\mathcal{T}^\pi V_w - V_w\|_{d^\pi}^2, \tag{7}$$

where Π is the projection onto $\{\Phi w : w \in \mathbb{R}^d\}$ in the d^π-norm (Lesson 51, Definition 4.3).

Remark 6.2 (Why MSPBE, Not MSBE?). The Mean Squared Bellman Error $\overline{\mathrm{BE}}(w) = \|\mathcal{T}^\pi V_w - V_w\|_{d^\pi}^2$ measures how far V_w is from satisfying the Bellman equation, but its minimizer is generally different from the TD fixed point. The MSPBE projects the Bellman error onto the representable subspace $\mathcal{V}$, and its minimizer is exactly the TD fixed point w_{TD} (where $\overline{\mathrm{PBE}}(w_{\mathrm{TD}}) = 0$).

6.1 Expressing the MSPBE in Tractable Form

Proposition 6.3 (Quadratic Form of the MSPBE)**.** *Under linear function approximation, the MSPBE can be written as*

$$\overline{\mathrm{PBE}}(w) = (Aw + b)^\top C^{-1}(Aw + b), \tag{8}$$

where $A = \Phi^\top D(\gamma P_\pi - I)\Phi$*,* $b = \Phi^\top D\, r_\pi$*, and* $C = \Phi^\top D\Phi$ *(the Gram matrix), with* $D = D^\pi$ *being the on-policy distribution.*

Proof. Since $V_w = \Phi w$, the Bellman residual is $\mathcal{T}^\pi V_w - V_w = r_\pi + \gamma P_\pi \Phi w - \Phi w$, and the projected residual is $\Pi(\mathcal{T}^\pi V_w - V_w)$. Using $\Pi = \Phi C^{-1}\Phi^\top D$ (Lesson 51, Definition 4.3):

$$\begin{aligned}\overline{\mathrm{PBE}}(w) &= \|\Pi(\mathcal{T}^\pi V_w - V_w)\|_D^2 = (\mathcal{T}^\pi V_w - V_w)^\top D\,\Pi^\top D\,\Pi(\mathcal{T}^\pi V_w - V_w)\\ &= (\mathcal{T}^\pi V_w - V_w)^\top D\,\Phi C^{-1}\Phi^\top D(\mathcal{T}^\pi V_w - V_w).\end{aligned} \tag{9}$$

Since $\Phi^\top D(\mathcal{T}^\pi V_w - V_w) = \Phi^\top D(r_\pi + \gamma P_\pi \Phi w - \Phi w) = b + Aw$, we have $\overline{\mathrm{PBE}}(w) = (Aw + b)^\top C^{-1}(Aw + b)$. □

Lemma 6.4 (Gradient of the MSPBE)**.** *The gradient of the MSPBE with respect to* w *is*

$$\nabla_w \overline{\mathrm{PBE}}(w) = 2A^\top C^{-1}(Aw + b). \tag{10}$$

Proof. Differentiating $(Aw+b)^\top C^{-1}(Aw+b)$ with respect to w: $\nabla_w = 2A^\top C^{-1}(Aw+b)$, since C^{-1} is symmetric positive definite and A is linear in w. □

Remark 6.5 (The C^{-1} Problem). Direct stochastic gradient descent on the MSPBE would require sampling an unbiased estimate of $\nabla_w \overline{\mathrm{PBE}}(w) = 2A^\top C^{-1}(Aw + b)$. The factor C^{-1} cannot be estimated from a single sample (it is an expectation of an outer product). This is the "double sampling" problem: a product of two expectations cannot be unbiasedly estimated from a single sample. GTD2 resolves this with a two-timescale approach.

6.2 The GTD2 Algorithm

Definition 6.6 (GTD2 Update Rules)**.** The *GTD2 algorithm* [Sutton et al., 2009] maintains two weight vectors $w \in \mathbb{R}^d$ (primary) and $\theta \in \mathbb{R}^d$ (secondary):

$$w_{t+1} = w_t + \alpha_t\big(\phi(S_t) - \gamma\,\phi(S_{t+1})\big)\phi(S_t)^\top\theta_t, \tag{11}$$

$$\theta_{t+1} = \theta_t + \beta_t\big(\delta_t - \phi(S_t)^\top\theta_t\big)\phi(S_t), \tag{12}$$

where $\delta_t = R_{t+1} + \gamma\, w_t^\top\phi(S_{t+1}) - w_t^\top\phi(S_t)$, and the step sizes satisfy $\alpha_t, \beta_t > 0$ with $\beta_t/\alpha_t \to \infty$ (the θ iteration runs on a faster timescale).

Remark 6.7 (Intuition for the Two Timescales). The secondary weight θ approximates $C^{-1}(Aw+b)$, which is the quantity needed for the MSPBE gradient. On the fast timescale, θ tracks the moving target $C^{-1}(Aw_t+b)$ for the slowly changing w_t. On the slow timescale, w_t updates using θ_t as a surrogate for $C^{-1}(Aw_t+b)$, effectively performing gradient descent on the MSPBE.

7 Convergence of GTD2

Assumption 7.1 (GTD2 Convergence Conditions).

1. Linear function approximation: $V_w(s) = w^\top \phi(s)$.
2. The feature matrix Φ has full column rank d.
3. The Markov chain under the behavior policy b is ergodic with stationary distribution d^b.
4. Step sizes: $\sum_t \alpha_t = \sum_t \beta_t = \infty$, $\sum_t \alpha_t^2 < \infty$, $\sum_t \beta_t^2 < \infty$, and $\alpha_t/\beta_t \to 0$.

Convergence of GTD2

Theorem 7.2 (Convergence of GTD2). *Under Assumption 7.1, the GTD2 iterates converge almost surely:*

$$(w_t, \theta_t) \xrightarrow{a.s.} (w_{\mathrm{TD}}, 0), \tag{13}$$

where $w_{\mathrm{TD}} = -A^{-1}b$ is the TD fixed point (the minimizer of $\overline{\mathrm{PBE}}$).

Proof. We verify the conditions of Borkar's two-timescale stochastic approximation theorem [Borkar, 2008] and apply it.

Step 1: Identify the two-timescale ODEs. The fast-timescale ODE (for θ, with w frozen) is obtained by computing the expected update:

$$\begin{aligned}\mathbb{E}[\text{increment of } \theta \mid w, \theta] &= \mathbb{E}[(\delta_t - \phi(S_t)^\top \theta)\phi(S_t)] = \mathbb{E}[\delta_t\, \phi(S_t)] - \mathbb{E}[\phi(S_t)\phi(S_t)^\top]\,\theta \\ &= (Aw+b) - C\,\theta,\end{aligned} \tag{14}$$

where $C = \Phi^\top D^b \Phi$ (using the behavior distribution d^b since states are sampled under b). The fast ODE is:

$$\dot{\theta} = (Aw+b) - C\,\theta. \tag{15}$$

Step 2: Verify fast ODE stability. The matrix $-C$ is negative definite (since $C = \Phi^\top D^b \Phi$ is positive definite by the full rank assumption and $d^b(s) > 0$ for all s). Therefore, for each fixed w, the fast ODE (15) is globally asymptotically stable with equilibrium

$$\theta^*(w) = C^{-1}(Aw+b). \tag{16}$$

Step 3: Derive the slow ODE. The slow-timescale ODE (for w, with $\theta = \theta^*(w)$) is:

$$\begin{aligned}\mathbb{E}[\text{increment of } w \mid w, \theta^*(w)] &= \mathbb{E}\big[(\phi(S_t) - \gamma\,\phi(S_{t+1}))\phi(S_t)^\top \theta^*(w)\big] \\ &= \mathbb{E}\big[(\phi(S_t) - \gamma\,\phi(S_{t+1}))\phi(S_t)^\top\big]\, C^{-1}(Aw+b) \\ &= -A^\top C^{-1}(Aw+b) = -\frac{1}{2}\nabla_w \overline{\mathrm{PBE}}(w),\end{aligned} \tag{17}$$

where we used $\mathbb{E}[(\phi(S_t) - \gamma\,\phi(S_{t+1}))\phi(S_t)^\top] = -A^\top$ (since $A = \mathbb{E}[\phi(S_t)(\gamma\,\phi(S_{t+1}) - \phi(S_t))^\top]$, so $A^\top = \mathbb{E}[(\gamma\,\phi(S_{t+1}) - \phi(S_t))\phi(S_t)^\top]$ and $-A^\top = \mathbb{E}[(\phi(S_t) - \gamma\,\phi(S_{t+1}))\phi(S_t)^\top]$). The slow ODE is:

$$\dot{w} = -A^\top C^{-1}(Aw+b) = -\frac{1}{2}\nabla_w \overline{\mathrm{PBE}}(w). \tag{18}$$

Step 4: Verify slow ODE stability. The MSPBE $\overline{\mathrm{PBE}}(w) = (Aw+b)^\top C^{-1}(Aw+b)$ is a non-negative quadratic form. Its gradient $\nabla_w \overline{\mathrm{PBE}}(w) = 2A^\top C^{-1}(Aw+b)$ vanishes if and

only if $Aw + b = 0$ (since C^{-1} is positive definite and A is invertible by the argument of Lesson 51 for the on-policy case; more generally, the MSPBE has a unique minimizer whenever A is invertible). The unique minimizer is $w_{\text{TD}} = -A^{-1}b$.

The Jacobian of the slow ODE at w_{TD} is $-A^\top C^{-1}A$, which is negative definite (since C^{-1} is positive definite and A is invertible). Therefore the slow ODE is globally asymptotically stable with equilibrium w_{TD}.

Step 5: Verify noise conditions. The noise terms in both updates are martingale differences with bounded conditional second moments (since all quantities are bounded on a finite MDP with bounded features). The step-size conditions in Assumption 7.1 match those required by Borkar's theorem.

Step 6: Apply the two-timescale theorem. By Borkar [2008, Theorem 2, Chapter 6], the coupled iteration (11)–(12) converges to the equilibrium of the coupled ODE system. Since the fast ODE has globally asymptotically stable equilibrium $\theta^*(w)$ for each w, and the slow ODE (with θ replaced by $\theta^*(w)$) has globally asymptotically stable equilibrium w_{TD}, we conclude:

$$(w_t, \theta_t) \to (w_{\text{TD}}, \theta^*(w_{\text{TD}})) = (w_{\text{TD}}, C^{-1}(Aw_{\text{TD}} + b)) = (w_{\text{TD}}, 0) \quad \text{a.s.} \tag{19}$$

The last equality uses $Aw_{\text{TD}} + b = 0$. □

Remark 7.3 (GTD2 Works Under Off-Policy Sampling). Crucially, the convergence proof does not require $d^b = d^\pi$. The fast ODE stability uses $C = \Phi^\top D^b \Phi$, which is positive definite for any distribution d^b with full support. The slow ODE performs gradient descent on the MSPBE, which is well-defined regardless of the sampling distribution. Thus GTD2 resolves the deadly triad by replacing the unstable semi-gradient update with a true gradient on a meaningful objective.

Example 7.4 (GTD2 on Baird's MDP)**.** Applying GTD2 to Baird's counterexample (Section 4) with step sizes $\alpha_t = 0.01/(t+1)^{0.6}$ and $\beta_t = 0.1/(t+1)^{0.5}$ (satisfying $\alpha_t/\beta_t \to 0$), the weights w_t converge to $w_{\text{TD}} = 0$ (the correct solution since $V^\pi \equiv 0$). The secondary weights θ_t converge to 0. In contrast, semi-gradient TD(0) with the same step size diverges.

8 Practical Resolutions of the Deadly Triad

Three Approaches to the Deadly Triad

Modern deep RL employs three main strategies:

1. **Target networks** (DQN): Freeze the bootstrap target $\gamma \max_{a'} Q_{\bar{w}}(S', a')$ using a periodically updated copy $\bar{w}$. This does not eliminate off-policy bootstrapping with FA but breaks the positive feedback loop by making the target stationary over intervals. No convergence guarantee exists for the nonlinear case.

2. **Gradient-TD methods** (GTD2, TDC): Replace the semi-gradient with a true gradient on the MSPBE. Provably convergent under linear FA (Theorem 7.2), but empirically slower than semi-gradient methods in deep RL due to the two-timescale learning and the secondary weight vector.

3. **Emphatic-TD**: Reweight updates by the *emphasis* M_t, a product of importance sampling ratios that corrects d^b back to d^π. Provably convergent under linear FA but can have high variance from the importance weights.

Proposition 8.1 (Target Networks Stabilize Semi-Gradient TD)**.** *Consider the modified update* $w_{t+1} = w_t + \alpha_t\big[R_{t+1} + \gamma\, \bar{w}^\top \phi(S_{t+1}) - w_t^\top \phi(S_t)\big]\phi(S_t)$*, where* $\bar{w}$ *is held fixed. This is equivalent*

to performing semi-gradient TD on a non-bootstrapped target, and converges for any sampling distribution.

Proof. With $\bar{w}$ fixed, the target $R_{t+1} + \gamma\, \bar{w}^\top \phi(S_{t+1})$ does not depend on the current weights w_t. The update becomes $w_{t+1} = w_t + \alpha_t[y_t - w_t^\top \phi(S_t)]\phi(S_t)$ where $y_t = R_{t+1} + \gamma\, \bar{w}^\top \phi(S_{t+1})$ is a regression target independent of w_t. This is stochastic gradient descent on $\mathbb{E}_{d^b}[(y - w^\top \phi(S))^2]$, a convex quadratic, which converges under standard conditions regardless of d^b. The bootstrapping element of the triad is effectively removed because the target is fixed. □

Remark 8.2 (Target Network Limitations). Target networks make the target stationary for a fixed interval, after which $\bar{w}$ is updated to w_t and the process repeats. This is not a single convex optimization but a sequence of them, and the overall procedure is not guaranteed to converge. In practice, soft target updates ($\bar{w} \leftarrow \tau w + (1-\tau)\bar{w}$ with $\tau \ll 1$) smooth the transitions and work well empirically.

9 Exercises

Exercise 9.1 (Verifying the Triad on a Two-State MDP). ($*$)
Consider two states $\{s_1, s_2\}$, one feature $\phi(s_1) = 1$, $\phi(s_2) = 2$, $\gamma = 0.9$, $r \equiv 0$. Target policy: always go to s_2. Behavior policy: uniform random.

(a) Compute $A_{\text{on}} = \Phi^\top D^\pi(\gamma P_\pi - I)\Phi$ and verify it is negative.
(b) Compute $A_{\text{off}} = \Phi^\top D^b(\gamma P_\pi - I)\Phi$ and determine its sign.
(c) If $A_{\text{off}} > 0$, confirm that semi-gradient TD diverges under d^b.

Exercise 9.2 (MSPBE Computation). ($*$)
For the MDP in Exercise 9.1, compute $\overline{\text{PBE}}(w)$ as a function of w and find w_{TD} by setting $\nabla_w \overline{\text{PBE}} = 0$. Verify that $V_{w_{\text{TD}}} = \Pi\, \mathcal{T}^\pi V_{w_{\text{TD}}}$.

Exercise 9.3 (GTD2 Expected Updates). ($**$)
For Baird's counterexample (Definition 4.1), compute the expected fast-timescale update $\mathbb{E}[(\delta_t - \phi(S_t)^\top \theta)\phi(S_t)]$ and the expected slow-timescale update $\mathbb{E}[(\phi(S_t) - \gamma\, \phi(S_{t+1}))\phi(S_t)^\top \theta]$ explicitly. Verify that the equilibrium $\theta^*(w) = C^{-1}(Aw + b)$ satisfies $\theta^*(0) = 0$ (since $b = 0$).

Exercise 9.4 (Distribution Mismatch Quantification). ($**$)
Consider a general MDP with on-policy distribution d^π and off-policy distribution d^b. Define the distribution mismatch ratio $\rho(s) = d^b(s)/d^\pi(s)$.

(a) Show that $A_{\text{off}} = \Phi^\top D^b(\gamma P_\pi - I)\Phi$ can be written as $A_{\text{off}} = \Phi^\top D^\pi \text{diag}(\rho)(\gamma P_\pi - I)\Phi$.
(b) Argue that when $\rho(s) \equiv 1$ (on-policy), $A_{\text{off}} = A$ and negative definiteness holds.
(c) Construct a simple example where ρ is highly non-uniform and A_{off} has a positive eigenvalue.

Exercise 9.5 (Target Network Convergence Analysis). ($***$)
Consider the target network update scheme: for $k = 0, 1, 2, \ldots$, fix $\bar{w} = w_{kN}$ for N steps, run semi-gradient TD to approximate convergence, then set $\bar{w} = w_{(k+1)N}$.

(a) Show that each block of N steps solves a convex regression problem with target $y_t = R_{t+1} + \gamma\, \bar{w}^\top \phi(S_{t+1})$.
(b) Let $w^*(\bar{w})$ denote the solution of this regression. Show that $w^*(\bar{w}) = \bar{w}$ if and only if $\bar{w} = w_{\text{TD}}$.
(c) Analyze the fixed-point iteration $\bar{w}_{k+1} = w^*(\bar{w}_k)$ and determine under what conditions it converges to w_{TD}.

Hint: The map $\bar{w} \mapsto w^*(\bar{w})$ is an affine contraction.

10 Summary

- The **deadly triad** — function approximation + bootstrapping + off-policy learning — can cause TD methods to diverge ($\|w_t\| \to \infty$), even in the simplest linear case with zero rewards.
- **Any two** elements of the triad are safe: on-policy TD with FA converges (Lesson 51), off-policy MC with FA converges, and tabular Q-learning converges off-policy (Lesson 50).
- **Baird's counterexample** provides a concrete 7-state MDP where semi-gradient TD(0) with linear FA diverges under off-policy sampling. The mechanism is a positive eigenvalue in $A_{\text{off}} = \Phi^\top D^b(\gamma P_\pi - I)\Phi$, caused by the mismatch $D^b \neq D^\pi$.
- The **root cause** is that the contraction $\|P_\pi v\|_{d^\pi} \leq \|v\|_{d^\pi}$ (which ensures A is negative definite) relies on the Jensen inequality argument with the *stationary distribution of* P_π. Under a different distribution d^b, this contraction fails.
- The **MSPBE** $\overline{\text{PBE}}(w) = (Aw+b)^\top C^{-1}(Aw+b)$ is a well-defined, non-negative objective whose minimizer is w_{TD}. Its gradient involves C^{-1}, which cannot be estimated from a single sample.
- **GTD2** resolves the deadly triad using a two-timescale iteration: the fast timescale learns $\theta \approx C^{-1}(Aw+b)$, and the slow timescale performs gradient descent on the MSPBE using θ as a surrogate. Convergence follows from Borkar's two-timescale theorem.
- **Target networks** provide a practical (but theoretically incomplete) resolution by freezing the bootstrap target, effectively removing bootstrapping for fixed intervals.
- The deadly triad is not merely a theoretical curiosity — it explains the instabilities observed in early DQN training and motivates the engineering solutions (target networks, replay buffers, gradient clipping) used in modern deep RL.

TD Learning Subdomain Summary

This lesson concludes the TD Learning subdomain (Lessons 47–53). The following table summarizes the progression.

Lesson	Topic	Key Result	Setting
47	TD(0) and n-step TD	TD(0) converges to V^π a.s.	Tabular, on-policy
48	TD(λ), eligibility traces	Forward–backward equivalence	Tabular, on-policy
49	SARSA (on-policy control)	SARSA converges to Q^* under GLIE	Tabular, on-policy
50	Q-learning (off-policy control)	Q-learning converges to Q^* a.s.	Tabular, off-policy
51	Linear function approximation	$\overline{\text{VE}}(w_{\text{TD}}) \leq \frac{1}{1-\gamma} \min_w \overline{\text{VE}}(w)$	Linear FA, on-policy
52	Approximate policy iteration	Error bound: $O(\varepsilon/(1-\gamma)^2)$	Linear FA, API
53	Deadly triad and GTD2	FA + bootstrap + off-policy can diverge; GTD2 converges via MSPBE gradient	Linear FA, off-policy

The arc traces a clear progression: from tabular prediction (47–48) to tabular control (49–50), then to function approximation (51–52), and finally to the fundamental limitations and resolutions (53). The central theme is the interplay between bootstrapping, function approximation, and distribution: when these align (on-policy, tabular), TD methods converge reliably; when they conflict (off-policy + FA + bootstrapping), careful algorithm design (GTD2, target networks) is required.

11 Further Reading

- **Phase 04 (Deep RL)** extends the deadly triad analysis to nonlinear function approximation. DQN (Lesson 30) uses target networks and experience replay as practical mitigations. Distributional RL (Lesson 31) encounters similar stability challenges.
- Baird [1995] introduced the counterexample and the concept of residual algorithms as an alternative resolution. Tsitsiklis and Van Roy [1997] provided the theoretical framework separating the on-policy and off-policy cases.
- Sutton et al. [2009] developed the GTD family (GTD, GTD2, TDC) and established their convergence properties. The TDC variant uses a different decomposition of the MSPBE gradient that can be empirically faster.
- Borkar [2008] provides the definitive treatment of two-timescale stochastic approximation, including the stability conditions and convergence rates used in the GTD2 proof.
- For emphatic-TD methods, see Sutton, Mahmood, and White (2016), which develops the emphatic weighting approach to restore convergence under off-policy sampling by importance weighting.
- The connection between the deadly triad and the instability of deep RL training is explored in van Hasselt et al. (2018), "Deep Reinforcement Learning and the Deadly Triad," which provides empirical evidence that all three elements contribute to instability in practice.

References

Leemon Baird. Residual algorithms: Reinforcement learning with function approximation. In *Proceedings of the 12th International Conference on Machine Learning*, pages 30–37, 1995.

Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, 1996.

Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.

Richard S. Sutton, Hamid Reza Maei, Doina Precup, Shalabh Bhatnagar, David Silver, Csaba Szepesvári, and Eric Wiewiora. Fast gradient-descent methods for temporal-difference learning with linear function approximation. *Proceedings of the 26th International Conference on Machine Learning*, pages 993–1000, 2009.

John N. Tsitsiklis and Benjamin Van Roy. An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5):674–690, 1997.